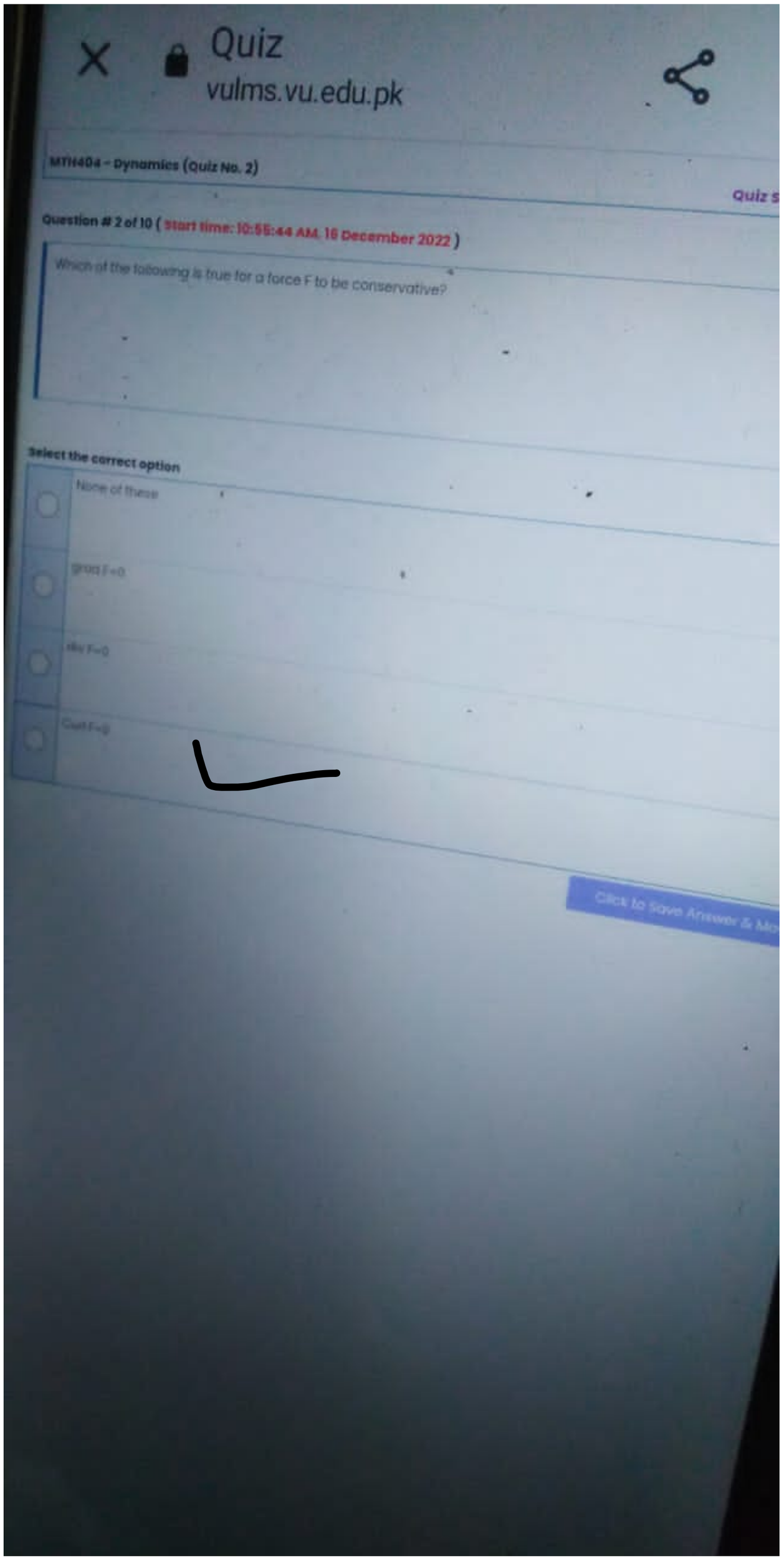
Question # 8 of 10 (Start time: 02:21:04 PM, 16 December 2022) **Total Marks:** 1

If force of magnitude 20N is applied to the particle and it covers a distance of 3m in the same direction as that of applied force, then the magnitude of work done on the particle is------.

Select the correct option

- ○ 3J
- ○ 60J
- ○ 20J
- ○ 23J

Click to Save Answer & Move to Next Question


Quiz
vulms.vu.edu.pk
MTH404 - Dynamics (Quiz No. 2)
Quiz s
Question # 2 of 10 (Start time: 10:55:44 AM, 16 December 2022)
Which of the following is true for a force F to be conservative?
Select the correct option
None of these
grad F=0
div F=0
Curl F=0
Click to save Answer & Mo

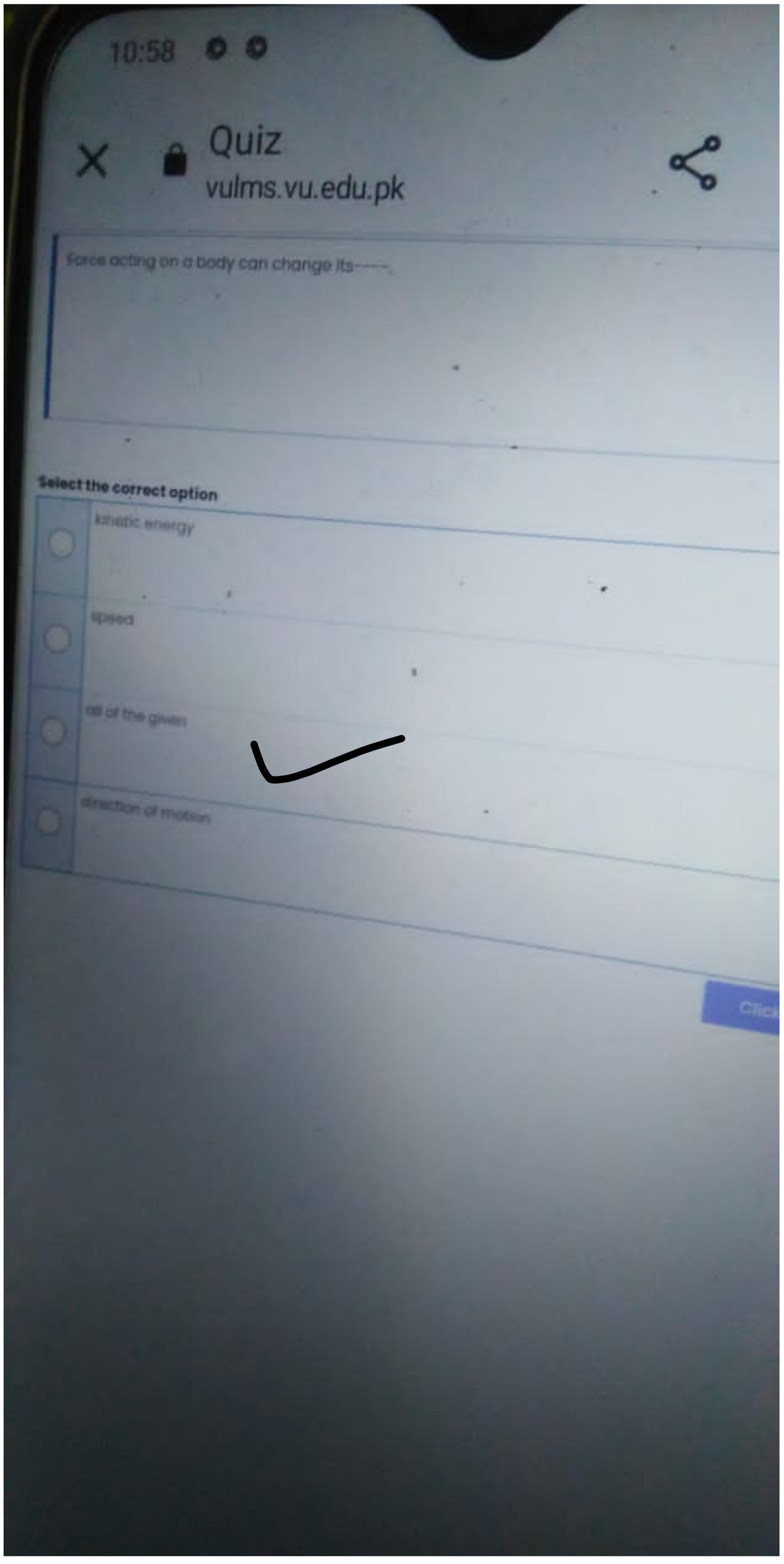
MTH404 - Dynamics (Quiz No. 2)

Question # 8 of 10 (Start time: 02:20:12 PM, 18 December 2022)

In a conservative field of force, the total energy of a particle------- throughout the motion.

Select the correct option

- none of these
- varies
- remains constant
- zero


10:58
Quiz
vulms.vu.edu.pk
Force acting on a body can change its-----.
Select the correct option
kinetic energy
speed
all of the given
direction of motion

10:56

Quiz

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Question # 3 of 10 (Start time: 10:56:22 AM, 16 December 2022)

If a particle of mass 4kg is moving with speed 3m/s2 then the kinetic energy of the particle is------

Select the correct option

- 12J
- 18J
- 36J
- 48J

Click to Save Answe

MTH404 - Dynamics (Quiz No. 2)

Time Left 80 sec(s)

Quiz Start Time: 02:13 PM, 16 December 2022

Question # 7 of 10 (Start time: 02:19:28 PM, 16 December 2022)

Total Marks:

If a force F acts at time t on a particle of mass m moving with velocity v, then the equation of motion, according to Newton's second law, is given by-------

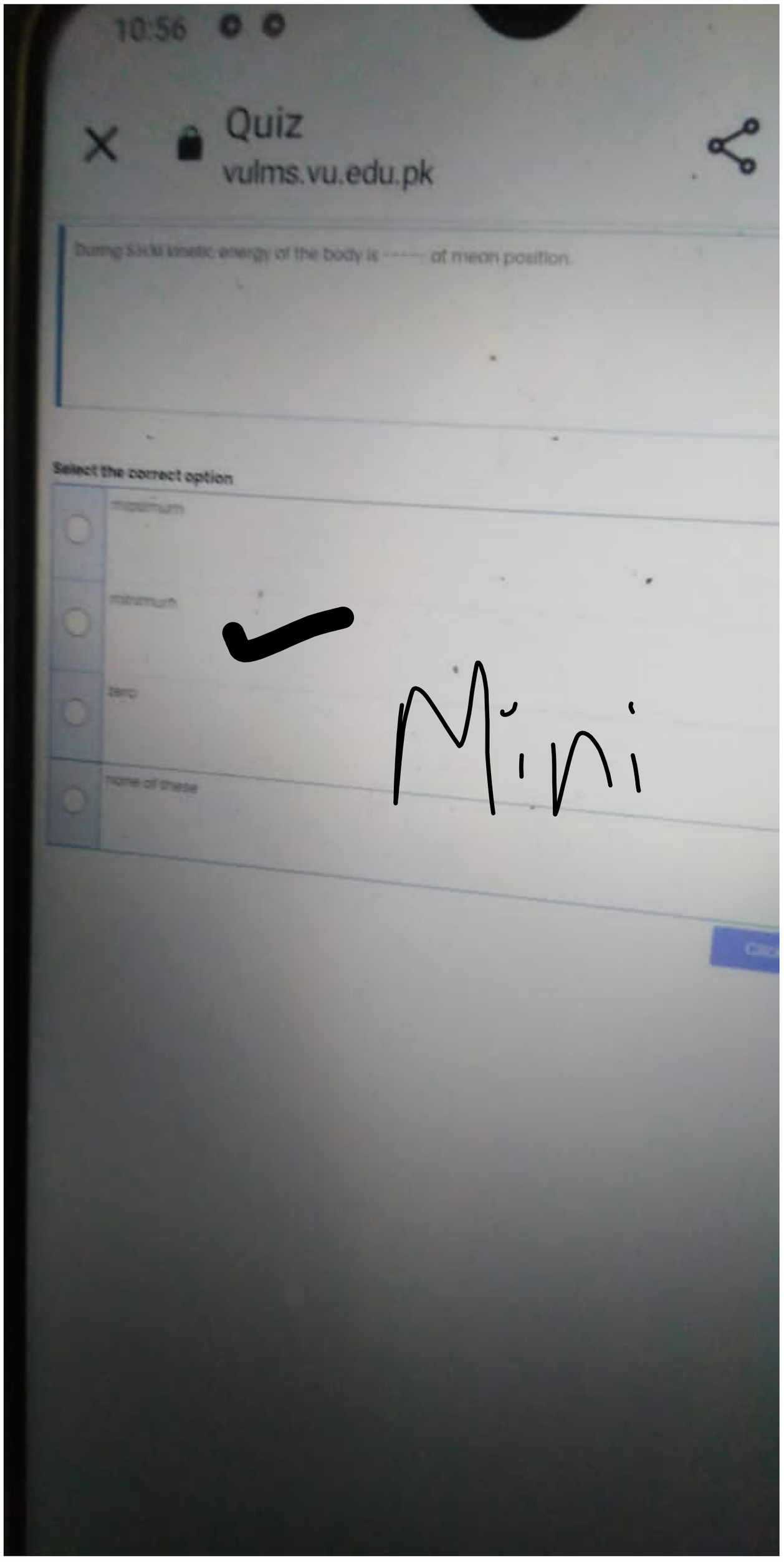
Select the correct option

Reload Math Equations

✓	$\vec{F} = \frac{d}{dt}(m\vec{v})$
	none of these
	$\vec{F} = m\vec{v}^2$
	$\vec{F} = m\vec{v}$

Click to Save Answer & Move to Next Question

Gohar Ch 2:22 PM 16-Dec-22


10:56
Quiz
vulms.vu.edu.pk
During SHM kinetic energy of the body is ------ at mean position.
Select the correct option
maximum
minimum
zero
none of these
Mini

Question # 9 of 10 (Start time: 02:22:28 PM, 16 December 2022) Total Marks: 1

Time period is reciprocal of-----.

Select the correct option

- amplitude
- frequency
- wavelength
- cycle

Click to Save Answer & Move to Next Question

Quiz

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MTH404 - Dynamics (Quiz No. 2)

Question # 8 of 10 (Start time: 10:58:37 AM, 16 December 2022)

What force can change the velocity of body of mass 1kg from 10m/s to 15m/s in one second?

Select the correct option

- 10N
- 1N
- 5N
- 15N

Question # 1 of 10 (Start time: 02:13:49 PM, 16 December 202

frequency is reciprocal of -----.

Select the correct option

- amplitude
- time period
- cycle
- wavelength

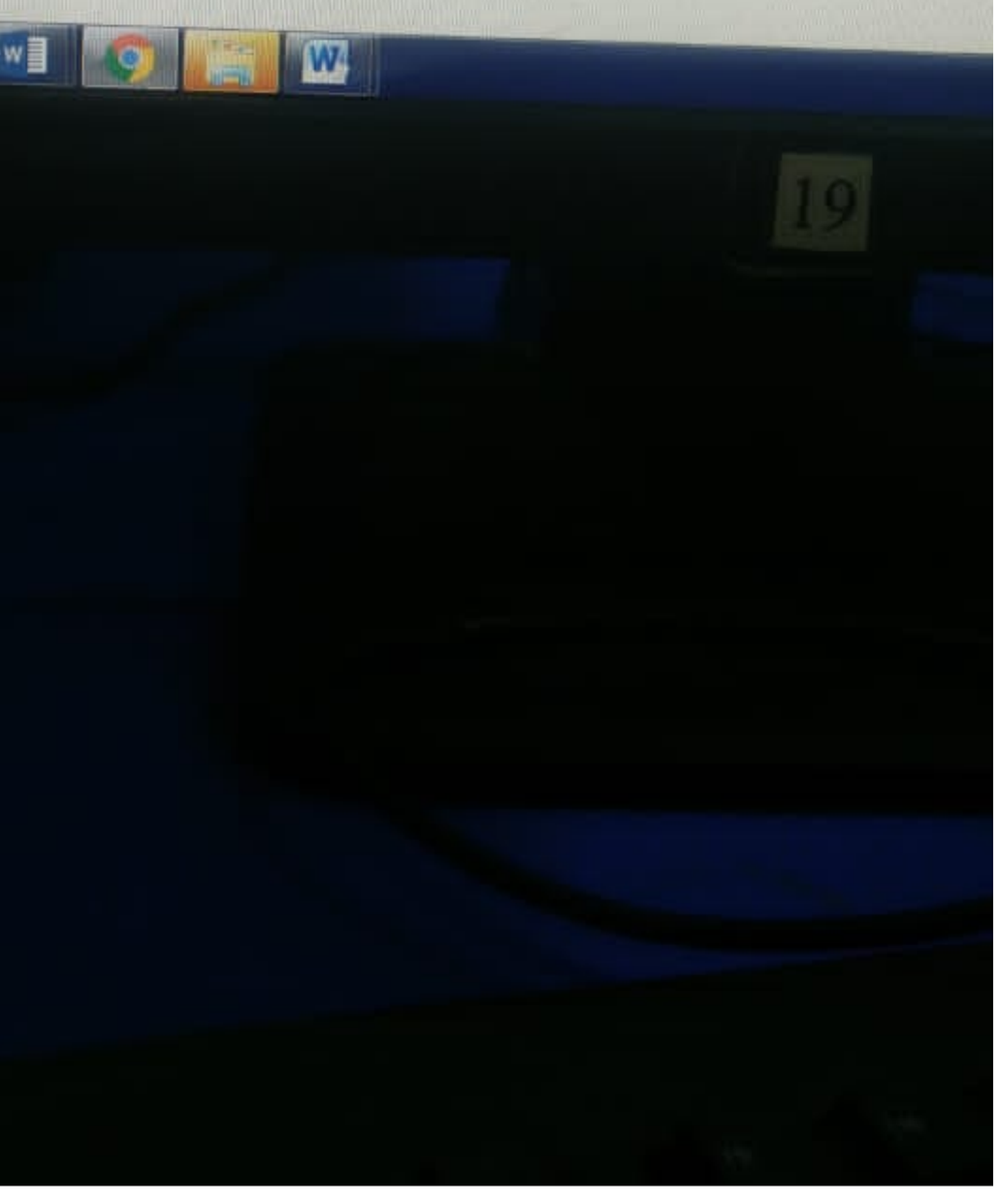
MTH404 - Dynamics (Quiz No. 2)

Question # 5 of 10 (Start time: 02:17:47 PM, 18 December 2022)

What force can change the velocity of body of mass 1kg from 10m/s to 15m/s in one second?

Select the correct option

- 10N
- 5N
- 1N
- 15N



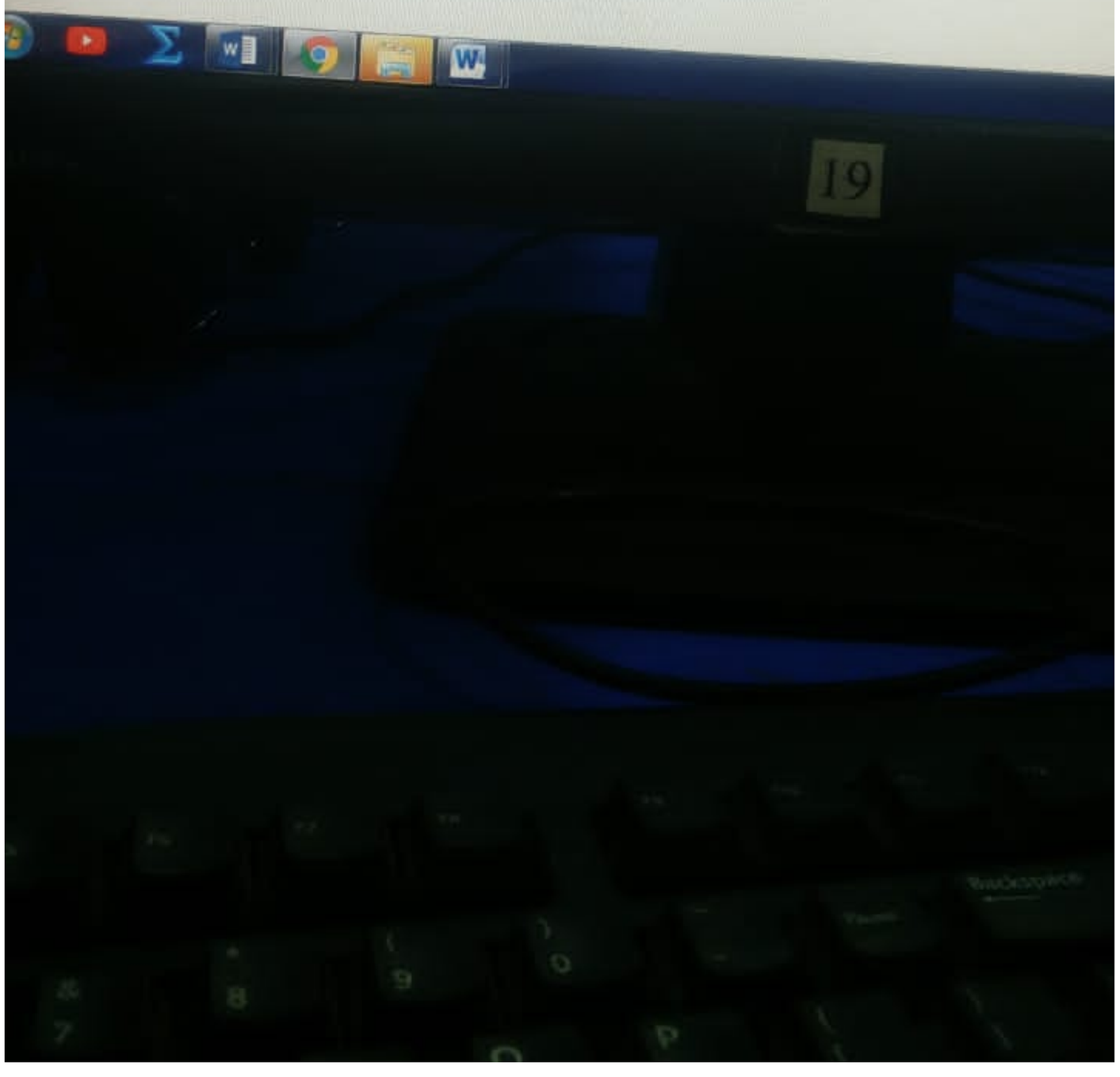
MTH404 - Dynamics (Quiz No. 2)

Question # 4 of 10 (Start time: 02:17:08 PM, 18 December 2022)

Which of the following is true about the transverse component of a force in polar coordinates?

Select the correct option

- $F_\theta = m(2\dot{r}\dot{\theta} + r\ddot{\theta})$
- $F_\theta = m(2\dot{r}\dot{\theta} - r\ddot{\theta})$
- $F_\theta = m(2\dot{r}\dot{\theta} + 2r\ddot{\theta})$
- None of thsese



Question # 6 of 10 (Start time: 02:13:16 PM, 16 December 2022) Total Marks: 1

A particle of mass 3kg is moving with velocity 4ti+5j. The force F acting on the particle will be----.

Select the correct option

- 4i+5j
- 12j
- 0
- 12i

Click to Save Answer & Move to Next Question

MTH404 - Dynamics (Quiz No. 2) Quiz Start Time: 02:06 PM, 16 December 2022

Question # 10 of 10 (Start time: 02:23:25 PM, 16 December 2022) **Total Marks:** 1

If a force F acts at time t on a particle of mass m moving with velocity v, then the equation of motion, according to Newton's second law, is given by-------.

Select the correct option

Reload Math Equations

○	$\vec{F} = m\vec{v}^2$
○	$\vec{F} = m\vec{v}$
○	none of these
○	$\vec{F} = \frac{d}{dt}(m\vec{v})$

Click to Save Answer & Move to Next Question

Question # 6 of 10 (Start time: 02:18:39 PM, 16 December 2022)

The rate of work done to the particle is known as-----.

Select the correct option

- none of these
- kinetic energy
- power
- momentum

...namics (Quiz No. 2)

Time Left 81 sec(s)

Question # 3 of 10 (Start time: 02:16:18 PM, 16 December 2022)

Quiz Start Time: 02:13 PM, 16 December 2022

Total Marks: 1

If a particle is moving along a circle of radius 'a' with Center at O, then which of the following is true about the transverse component of a force in polar coordinates?

Select the correct option

Reload Math Equations

- $F_\theta = ma\theta$
- $F_\theta = ma\ddot{\theta}$
- None of these
- $F_\theta = m\theta$

Click to Save Answer & Move to Next Question

Gohar Ch 2:19 PM 16-Dec-22

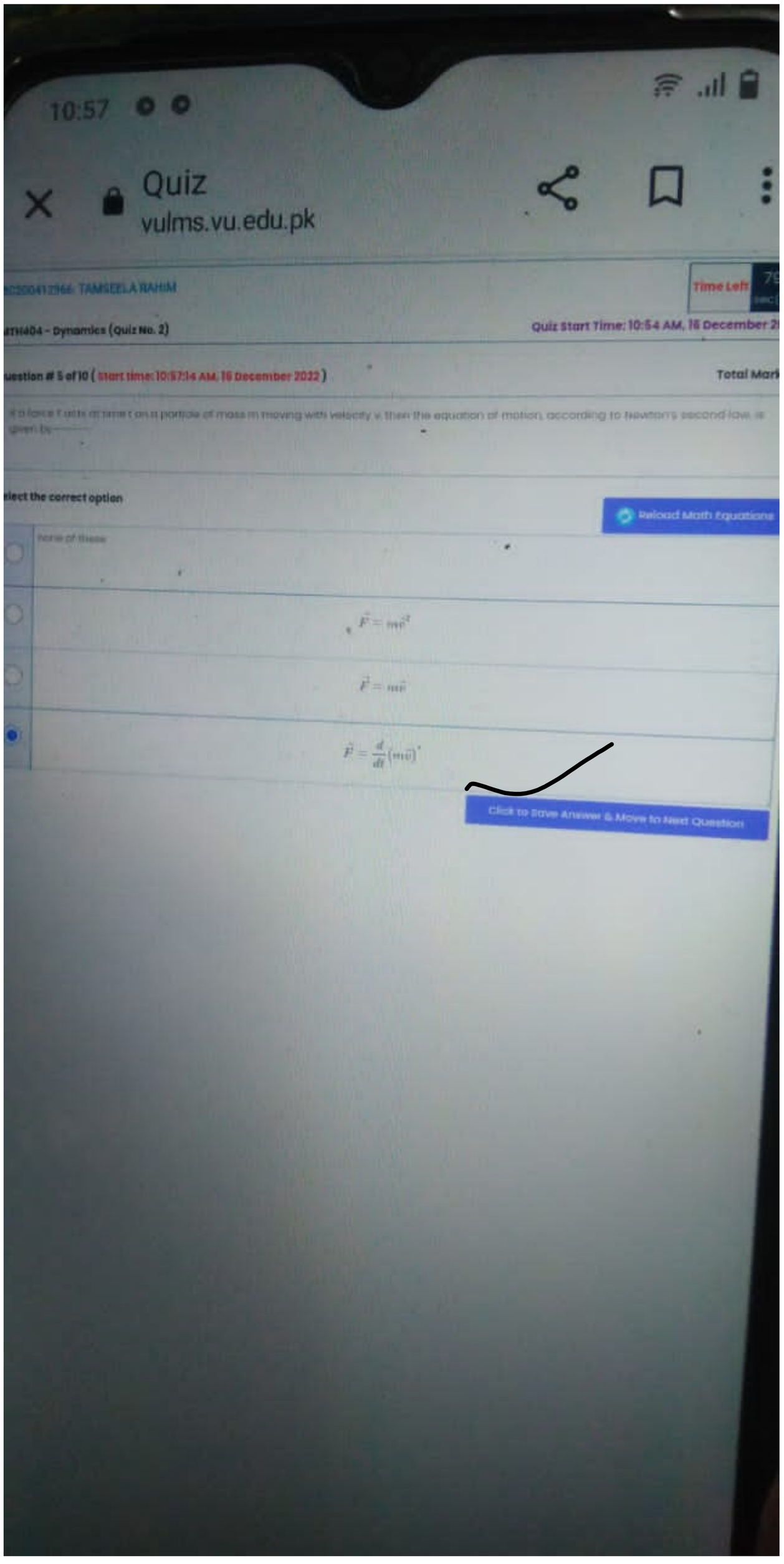
Question # 7 of 10 (Start time: 02:20:10 PM, 16 December 2022) Total Marks: 1

The rate of work done to the particle is known as-----.

Select the correct option

- none of these
- kinetic energy
- momentum
- power

Click to Save Answer & Move to Next Question

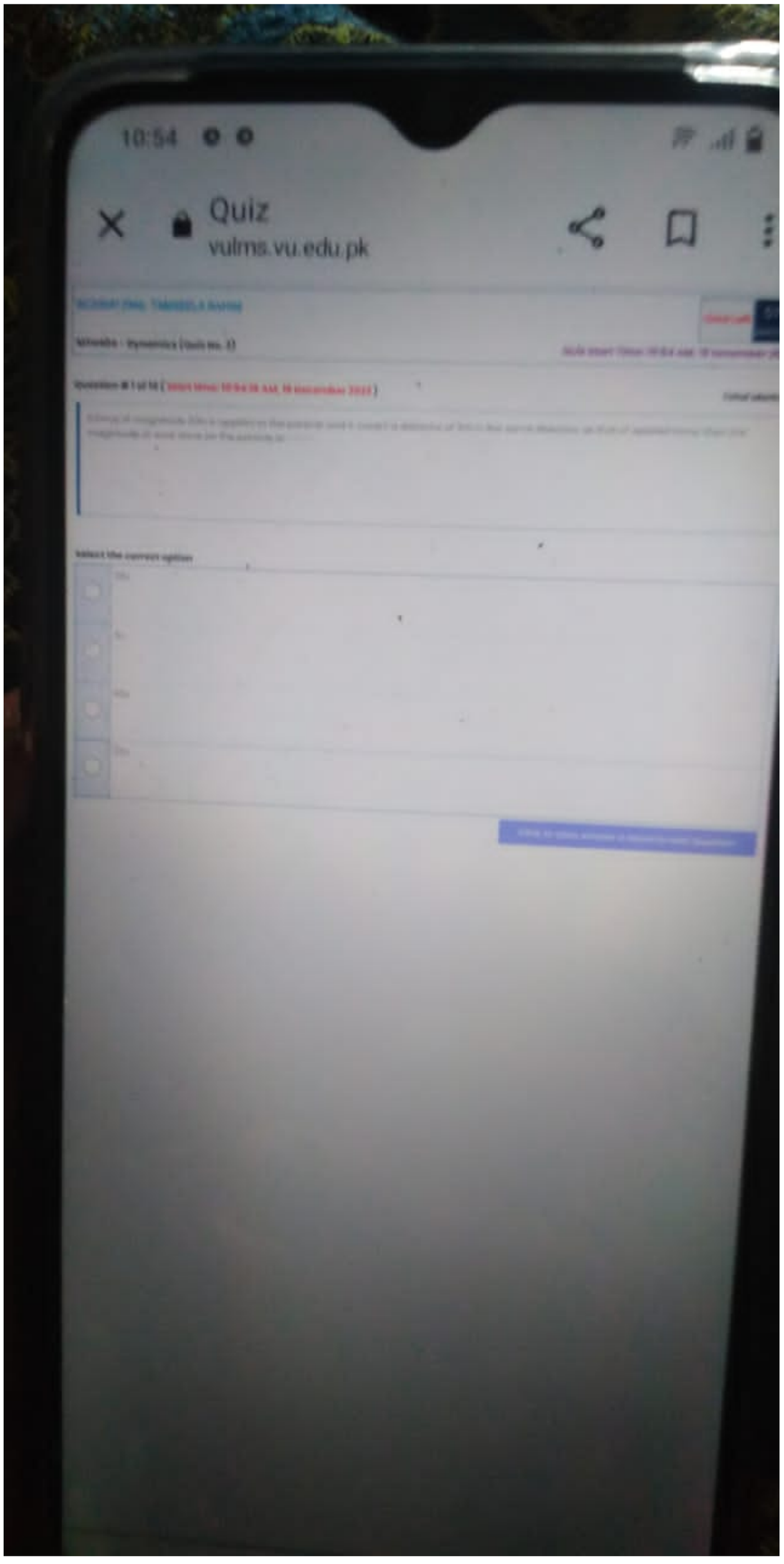

10:57
Quiz
vulms.vu.edu.pk
Time Left
Quiz Start Time: 10:54 AM, 16 December 2
Total Mark
Reload Math Equations
none of these
$\vec{F} = m\vec{v}^2$
$\vec{F} = m\vec{a}$
$\vec{F} = \frac{d}{dt}(m\vec{v})$
Click to Save Answer & Move to Next Question

Question # 10 of 10 (Start time: 02:21:20 PM, 16 December 2022)

A particle of mass 3kg is moving with velocity 4ti+5j. The force F acting on the particle will be----.

Select the correct option

- 4i+5j
- 12i
- 0
- 12j


10:54
Quiz
vulms.vu.edu.pk

Question # 5 of 10 (Start time: 02:12:14 PM, 16 December 2022) Total Marks: 1

If a particle of mass 4kg is moving with speed 3m/s2 then the kinetic energy of the particle is-----.

Select the correct option

- 48J
- 36J
- 12J
- 18J

Click to Save Answer & Move to Next Question

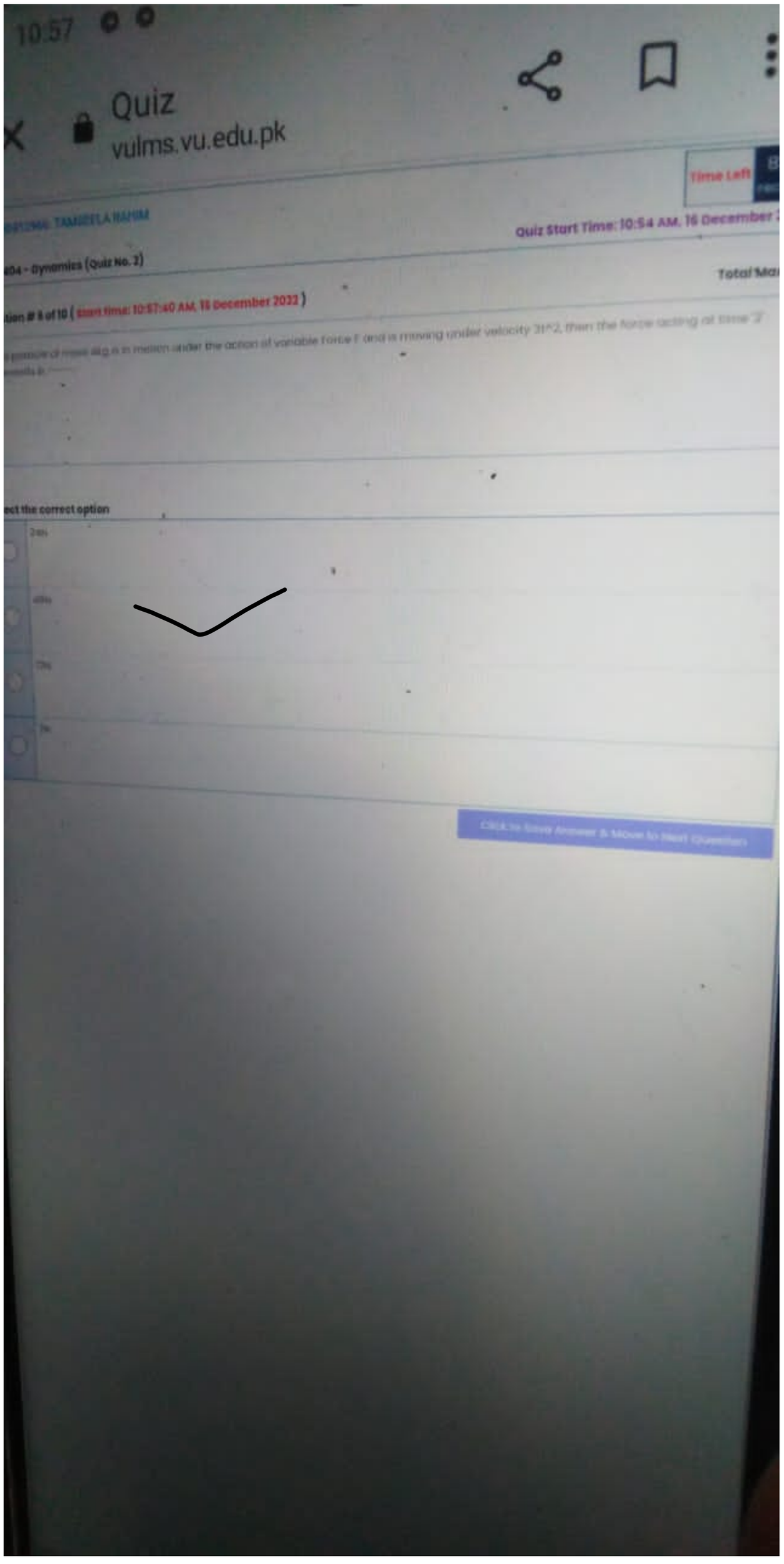
MTH404 - Dynamics (Quiz No. 2)

Question # 9 of 10 (Start time: 02:20:44 PM, 16 December 2022)

Time period is reciprocal of-----.

Select the correct option

- frequency
- wavelength
- amplitude
- cycle


10:57
Quiz
vulms.vu.edu.pk
Time Left
TAMEEELA RAHIM
Quiz Start Time: 10:54 AM, 16 December
04 - Dynamics (Quiz No. 2)
Total Ma
ion # 8 of 10 (start time: 10:57:40 AM, 16 December 2022)
ect the correct option
2ms
4ms
Click to Save Answer & Move to Next Question

vulms.vu.edu.pk/Qu

BC200403468: HIZB E ALI KHAN

Time Left 38 sec(s)

MTH404 - Dynamics (Quiz No. 2)

Quiz Start Time: 04:47 PM, 15 December 2022

Question # 2 of 10 (Start time: 04:48:10 PM, 15 December 2022)

Total Marks: 1

Which of the following is true about the transverse component of a force in polar coordinates?

Select the correct option

Reload Math Equations

- $F_\theta = m(2\dot{r}\dot{\theta} + 2r\ddot{\theta})$
- None of thsese
- $F_\theta = m(2\dot{r}\dot{\theta} - r\ddot{\theta})$
- $F_\theta = m(2\dot{r}\dot{\theta} + r\ddot{\theta})$ ✓

Saving...

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BC200403468: HIZB E ALI KHAN

Time Left 52 sec(s)

MTH404 - Dynamics (Quiz No. 2)

Quiz Start Time: 04:47 PM, 15 December 2022

Question # 10 of 10 (Start time: 04:53:55 PM, 15 December 2022)

Total Marks: 1

If a particle is moving along a circle of radius 'a' with Center at O, then which of the following is true about the transverse component of a force in polar coordinates?

Select the correct option

Reload Math Equations

○	$F_\theta = ma\theta$
○	None of these
●	$F_\theta = ma\ddot{\theta}$
○	$F_\theta = m\theta$

Click to Save Answer & Move to Next Question

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BC200403468: HIZB E ALI KHAN

Time Left 53 sec(s)

MTH404 - Dynamics (Quiz No. 2)

Quiz Start Time: 04:47 PM, 15 December 2022

Question # 4 of 10 (Start time: 04:50:12 PM, 15 December 2022)

Total Marks: 1

If a force F acts at time t on a particle of mass m moving with velocity v, then the equation of motion, according to Newton's second law, is given by-------.

Select the correct option

Reload Math Equations

- $\vec{F} = m\vec{v}$
- $\vec{F} = m\vec{v}^{2}$
- none of these
- $\vec{F} = \frac{d}{dt}(m\vec{v})$

Click to Save Answer & Move to Next Question

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BC200403468: HIZB E ALI KHAN

Time Left 64 sec(s)

MTH404 - Dynamics (Quiz No. 2)

Quiz Start Time: 04:47 PM, 15 December 2022

Question # 9 of 10 (Start time: 04:53:22 PM, 15 December 2022)

Total Marks: 1

Which of the following is true about the radial component of a force in polar coordinates?

Select the correct option

Reload Math Equations

- None of these
- $F_r = m(\ddot{r} + r(\dot{\theta})^2)$
- $F_r = m(r(\dot{\theta})^2 - \ddot{r})$
- $F_r = m(\ddot{r} - r(\dot{\theta})^2)$ ✓

Click to Save Answer & Move to Next Question

vulms.vu.edu.pk/Qu

BC200403468: HIZB E ALI KHAN

Time Left 48 sec(s)

MTH404 - Dynamics (Quiz No. 2)

Quiz Start Time: 04:47 PM, 15 December 2022

Question # 3 of 10 (Start time: 04:49:04 PM, 15 December 2022)

Total Marks: 1

What force can change the velocity of body of mass 1kg from 10m/s to 15m/s in one second?

Select the correct option

- 10N
- 1N
- 5N
- 15N

Click to Save Answer & Move to Next Question

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BC200403468: HIZB E ALI KHAN

Time Left 37 sec(s)

MTH404 - Dynamics (Quiz No. 2)

Quiz Start Time: 04:47 PM, 15 December 2022

Question # 6 of 10 (Start time: 04:51:06 PM, 15 December 2022)

Total Marks: 1

If a particle of mass 4kg is moving with speed 3m/s2 then the kinetic energy of the particle is-----.

Select the correct option

- 48J
- 12J
- 36J
- 18J

Click to Save Answer & Move to Next Question

vulms.vu.edu.pk/Qu

BC200403468: HIZB E ALI KHAN

Time Left 54 sec(s)

MTH404 - Dynamics (Quiz No. 2)

Quiz Start Time: 04:47 PM, 15 December 2022

Question # 1 of 10 (Start time: 04:47:18 PM, 15 December 2022)

Total Marks: 1

If Force of magnitude 10N is acting on the particle of mass 5kg, then the acceleration produced in the particle is-----.

Select the correct option

- 10m/s2
- 15m/s2
- 2m/s2
- 5m/s2

Click to Save Answer & Move to Next Question

vulms.vu.edu.pk/Qu

BC200403468: HIZB E ALI KHAN

Time Left 83 sec(s)

MTH404 - Dynamics (Quiz No. 2)

Quiz Start Time: 04:47 PM, 15 December 2022

Question # 7 of 10 (Start time: 04:52:04 PM, 15 December 2022)

Total Marks: 1

Time period is reciprocal of-----.

Select the correct option

- wavelength
- amplitude
- frequency
- cycle

Click to Save Answer & Move to Next Question

vulms.vu.edu.pk/Qu

BC200403468: HIZB E ALI KHAN

Time Left 48 sec(s)

MTH404 - Dynamics (Quiz No. 2)

Quiz Start Time: 04:47 PM, 15 December 2022

Question # 3 of 10 (Start time: 04:49:04 PM, 15 December 2022)

Total Marks: 1

What force can change the velocity of body of mass 1kg from 10m/s to 15m/s in one second?

Select the correct option

- 10N
- 1N
- 5N
- 15N

Click to Save Answer & Move to Next Question

vulms.vu.edu.pk/Qu

BC200403468: HIZB E ALI KHAN

Time Left 82 sec(s)

MTH404 - Dynamics (Quiz No. 2)

Quiz Start Time: 04:47 PM, 15 December 2022

Question # 5 of 10 (Start time: 04:50:53 PM, 15 December 2022)

Total Marks: 1

frequency is reciprocal of -----.

Select the correct option

- cycle
- wavelength
- amplitude
- time period

Click to Save Answer & Move to Next Question

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BC200403468: HIZB E ALI KHAN

Time Left 30 sec(s)

MTH404 - Dynamics (Quiz No. 2)

Quiz Start Time: 04:47 PM, 15 December 2022

Question # 8 of 10 (Start time: 04:52:16 PM, 15 December 2022)

Total Marks: 1

If a particle of mass 4kg is in motion under the action of variable Force F and is moving under velocity 3t^2, then the force acting at time '2' seconds is -----.

Select the correct option

- (●) 48N
- ( ) 7N
- ( ) 24N
- ( ) 12N

Click to Save Answer & Move to Next Question